8.5.1 Median-Ungrouped Frequency Distribution

(i) Arrange the data in ascending (or) decending order of magnitude.

(ii) Construct the cumulative frequency distribution. Let N be the total frequency.

(iii) If N is odd, median $= \left(\frac{N+1}{2}\right)^{th}$ observation.

(iv) If N is even, median $= \left(\frac{\left(\frac{N}{2}\right)^{\text{th}} \text{ observation} + \left(\frac{N}{2}+1\right)^{\text{th}} \text{ observation}}{2}\right)$

Example 8.11 Calculate the median for the following data:

Height (cm)	160	150	152	161	156	154	155
No. of Students	12	8	4	4	3	3	7

Solution

Let us arrange the marks in ascending order and prepare the following data:

Height (cm)	Number of students (f)	Cumulative frequency (cf)
150	8	8
152	4	12
154	3	15
155	7	22
156	3	25
160	12	37
161	4	41

Here N = 41

Median = size of $\left(\frac{N+1}{2}\right)^{th}$ value = size of $\left(\frac{41+1}{2}\right)^{th}$ value = size of 21st value.

If the 41 students were arranged in order (of height), the 21st student would be the middle most one, since there are 20 students on either side of him/her. We therefore need to find the height against the 21st student. 15 students (see cumulative frequency) have height less than or equal to 154 cm. 22 students have height less than or equal to 155 cm. This means that the 21st student has a height 155 cm.

Therefore, Median = 155 cm

8.5.2 Median - Grouped Frequency Distribution

In a grouped frequency distribution, computation of median involves the following

Steps

(i) Construct the cumulative frequency distribution.

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